DEPARTMENT OF COMPUTER APPLICATIONS
PSG COLLEGE OF TECHNOLOGY
MCA G1 & G2, SEMESTER 1
26MX16 - PROGRAMMING AND DATA STRUCTURES LABORATORY
PRACTICE SHEET 4

Prerequisite	**String traversal, character checking, pattern matching, searching, counting, position tracking, word processing, parsing, comparison and classification.**

1. An IT company's employee portal requires every employee to have a unique username. Write a C program to validate a username according to the following rules:
   - Length must be between 6 and 15 characters.
   - The first character must be an alphabet.
   - Only alphabets, digits, and underscore _ are allowed.
   - Spaces are not allowed.
   - Display the specific reason when the username is invalid.

   Example:
   Input : arun_123
   Output : Valid Username

2. A company's authentication system evaluates password strength before allowing account creation. Write a C program to classify a password as:
   - Weak
   - Moderate
   - Strong

   Rules:
   - Minimum length: 8 characters.
   - At least one uppercase letter.
   - At least one lowercase letter.
   - At least one digit.
   - At least one special character.

   The program should also report which security requirements are missing.
   Example:
   Input : Hello123
   Output : Moderate Password
   Missing : Special Character

3. An enterprise application collects employee email IDs during registration. Develop a C program to validate an email address using the following business rules:
   - Exactly one @ must be present.
   - The username must not be empty.
   - A domain must exist after @.
   - The domain must contain ..
   - Spaces are not allowed.
   - @ cannot be the first or last character.
   - . cannot immediately follow @.

   Example:
   Input : student@psgtech.ac.in
   Output : Valid Email

4. A DevOps team receives server logs containing messages such as:
   INFO Server started
   ERROR Database connection failed
   WARNING Memory usage high
   ERROR Authentication failed

   Write a C program to analyze the log text and determine:
   - Number of INFO messages
   - Number of WARNING messages
   - Number of ERROR messages

   Also display:
   System Status: NORMAL
   if there are no errors; otherwise display:
   System Status: ATTENTION REQUIRED

5. A search engine needs to determine how frequently a keyword appears in a document. Write a C program that accepts:
   - A document/text
   - A search keyword

   The program should:
   - Search for the keyword.
   - Count its occurrences.
   - Perform the search without considering uppercase/lowercase differences.
   - Display the positions where the keyword occurs.

   Example:
   Text : C programming is powerful. Learning C programming is useful.
   Keyword : programming
   Occurrences : 2

6. A cybersecurity system scans incoming messages for suspicious words. Develop a C program that scans a message for the following keywords:
   malware
   phishing
   ransomware
   attack
   virus
   The program should:
   - Search the message case-insensitively.
   - Identify all suspicious keywords.
   - Count the number of detected keywords.

   Classify the message as:
   SAFE
   SUSPICIOUS
   HIGH RISK
   Suggested rule:
   - 0 matches → SAFE
   - 1–2 matches → SUSPICIOUS
   - More than 2 → HIGH RISK

7. A software deployment system needs to determine whether an update is newer than the currently installed version.
   Given:
   Current Version : 2.9.10
   New Version : 2.10.3
   Write a C program to compare the two version strings and display:
   New Version is OLDER
   New Version is SAME
   New Version is NEWER
   Important:
   The program must compare numeric components, not ASCII characters.
   For example:
   2.10.3 > 2.9.10
   Even though character-by-character comparison would produce a different result.

8. A data transmission system wants to reduce the size of strings containing repeated characters. Implement a simple Run-Length Encoding (RLE) algorithm.
   Convert:
   aaabbccccdaa
   into:
   a3b2c4d1a2
   The program should:
   - Read a string.
   - Identify consecutive repeated characters.
   - Store each character followed by its frequency.

- Handle characters occurring only once.

Challenge:
Do not use a separate library function for compression.

9. A banking application should not display complete sensitive information on the screen. Write a C program to mask sensitive information.
   For example:
   Input:
   9876543210
   Output:
   ******3210
   For an email:
   kumar@gmail.com
   the output could be:
   *****@gmail.com
   The program should provide masking based on the type of input.
   Additional constraint:
   The original string must not be displayed after masking.

10. An IT company wants to build a simple text analytics module for analyzing customer feedback. Develop a C program that accepts a paragraph and generates the following report:
    TEXT ANALYTICS REPORT

    Total Characters :
    Total Words :
    Longest Word :
    Shortest Word :
    Unique Words :
    Repeated Words :
    Most Frequent Word:
    The program should:
    - Count characters.
    - Count words.
    - Identify the longest word.
    - Identify the shortest word.
    - Find unique words.
    - Identify repeated words.
    - Calculate word frequencies.
    - Display the most frequently occurring word.

    Example:
    Input:
    C programming is easy and C programming is powerful

    Output:
    Total Words : 9
    Longest Word : programming
    Unique Words : 6
    Repeated Words : C, programming, is
    Most Frequent Word: programming
    Frequency : 2

Skills Acquired	**String validation, pattern detection, keyword searching, frequency analysis, word analysis, string parsing, position tracking and rule based classification.**